

ChainLock

Hash-chain lattice memory for aziel-runtime

CL-WP-0.4 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Not a public GitHub release. Not a fleet-completeness claim. Runtime: `chainlock.py` · Schema: `chainlock_schema.json` · Namespace: `chainlock@azel`

0. Sentence

An AI on this runtime forgets at the session boundary. The lattice does not. ChainLock is the walkable memory between those two facts.

1. Claim

A local set of append-only chains, cross-linked as a lattice and optionally signed with an SSH key, is enough for an MCP-facing model to reconstruct short-term and long-term recollection without an embedding store and without rewriting history.

GodLock already hash-chains a public node. TemporalLock already timestamps receipts. FragGate already appends an act ledger. AZL-WP-1.1 already set UI = MCP. None of those give a reader a depth knob. ChainLock does.

This paper is the doctrine. The spec is `ChainLock_v0_spec.md`. The kit is the runtime. Recognition is not a third kit.

2. What it is not

- Not a public ledger. GodLock remains the public mesh stress-test node at `godlock.uk`.
- Not a substitute for TemporalLock receipts or EmbryoLock vaults. Those stay the evidence and sealed-file tools. ChainLock indexes and walks them.
- Not an embedding store. Match is deterministic: tags, subject, chain, time, hash, fact text.
- Not an untraceable path. SSH stamps identify a local key. Hashes identify bytes.
- Not FoldLock, SweepGate, or AZPIPE. Those are the wire. This is the memory.
- Not Unowned Lattice. UL papers stay UL.
- Not a court filing and not a prophetic document.

3. Units

Stamp. One JSON object. Canonical bytes are UTF-8 JSON with sorted keys and separators (`'`, `,`, `:`). `stamp_sha256` is SHA-256 of that encoding with the hash field omitted. `prev` is the previous stamp's hash on the same chain, or GENESIS for sequence zero.

Card. The wire unit. Slim keys: `id`, `c`, `k`, `t`, `h`, `fh`, `s`, `f`, `g`, `r?`. Extra `facts[]` only on capsules. Subject ≤ 80 characters. Fact ≤ 160 . Cap 4096 bytes. Hash-only cards are non-compliant. Recollection is the fact text. Grounding is `h` (stamp bytes) plus `fh` (SHA-256 of the clipped fact).

Chain. A single hash list stored as jsonl. Default roster: `genesis`, `identity`, `ssh`, `session`, `acts`, `evidence`, `recall`, `mesh`, `library`, `learn`.

Lattice. Typed links between stamps (cites, window-start, born-from, from-interact, quarantine). Links do not mutate the target.

SSH stamp. Detached ssh-keygen -Y sign over the canonical stamp bytes in namespace chainlock@aziel. Fingerprint lives on the identity chain. Unsigned stamps are chain-valid and not identity-grade.

Horizon. stm is the hot window. ltm is retained. both is counted in both. Compaction writes a recall capsule. Capsules hash the ordered member hashes and store the member facts. Source stamps stay.

Depth. Six classes. 0 tip. 1 STM. 2 capsules. 3 expand sources. 4 one lattice hop. 5 genesis or budget. auto starts at STM and deepens until a query hits or the budget dies.

4. Recollection

Short-term: last 16–32 session and acts stamps, plus the live tips.

Long-term: recall capsules first. Expand a capsule only when the question needs a source stamp. Follow one lattice hop only when depth ≥ 4 .

This is the same progressive-disclosure rule the skill loader already uses. Metadata, then body, then the bound file. ChainLock applies it to time.

Every returned fact carries id + h + fh. No claim without a stamp. FragGate door chainlock__door returns a grounded FG-CL envelope or refuse=no-stamp. Do not invent a fact to fill the door.

5. Library tether and adapt

Origin: <https://www.azielcorpuslibrary.net>.

The library chain holds catalog cards only — AZDOC id, title, content_sha256, document chain_tip, receipt URL. PDF bodies stay on the site. vault/library.jsonl is the offline shelf. If the Worker dies or a record is deleted from the public catalog, recall still has the card.

tether binds the current SSH identity to the library window hash. One stamp covers the set. New syncs require a new tether. Do not invent AZDOC ids or file hashes. Sync from /v1/search. Quarantined rows stay off the shelf.

Every operator action or choice gets an interact stamp on session. choose / refuse / -- learn also write learn. Adaptive recall is that chain plus its capsules. This is not a weight file and not a fine-tune.

6. MCP

The operator surface is tools, not a page.

```
chainlock__status chainlock__append chainlock__walk chainlock__recall chainlock__verify
chainlock__compact chainlock__door chainlock__mesh_cite chainlock__library_sync
chainlock__tether chainlock__interact chainlock__seal chainlock__pipe
```

Local process. No public panel. No IP allow/block UI. No Node Gate.

7. LOCKSET

seal writes three things that must agree:

1. Every live chain tip (id, h, fh). 2. A TemporalLock receipt: neutral timestamp, no narrative, no authority. 3. A GodLock cite: <https://godlock.uk>, pull=false, isolate=true.

verify is fail-closed. One broken prev, one receipt hash miss, one missing GodLock cite, or one tip drift after seal returns `lattice:false`. Append, interact, or library-sync after seal drifts tips. Reseal.

This runtime cites GodLock. It does not write the public ledger. The operator posts the lockset SHA-256.

8. Failure and poison

Broken prev or mismatched `stamp_sha256` isolates that chain. Do not rewrite. Append a verify stamp recording the break hash.

Failed SSH verify flags `SSH_FAIL`. The stamp stays.

Poison or unexpected remote citation records on mesh as `rel=quarantine`. SweepGate airlocks the inbound envelope before entry. Remote body does not enter STM.

Phoenix / ARK re-seal of a bound file does not edit the old stamp. Append a new pointer.

Court-sealed is a plane tag. It does not encrypt. Use ARK / EmbryoLock for ciphertext. Do not put legal name, home, passwords, or privileged plaintext in a stamp body.

9. Files

```
vault/
  chains/<name>.jsonl
  cards.jsonl
  library.jsonl
  receipts/LOCKSET.json
  receipts/TL-<prefix>.json
  keys/allowed_signers
  sigs/<stamp-id>.sig
  index.json
```

`index.json` and `cards.jsonl` are derived. Source of truth is the jsonl chains. Private SSH keys do not live on the artifacts FUSE mount.

10. Cap

This does not make the model omniscient. It makes forgetting an explicit depth choice instead of an accident.

Compat reads CL-0.1 through CL-0.4. New stamps are CL-0.4.

Companion papers: AP-WP-0.2 (wire), SG-WP-0.1 (airlock), LS-WP-0.1 (fail-closed seal).

AZPIPE

Fold-locked API path for aziel-runtime

AP-WP-0.2 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Not a fleet-completeness claim. Runtime: `azpipe.py` · Magic: FLD3 · Version: AZPIPE-0.2

0. Sentence

All data on the API, in and out, travels a fixed hop list. Architecture is visible on the envelope. Memory and tools never see raw inbound bytes.

1. Claim

A Worker, a library call, a mesh cite, and an MCP tool response are the same class of object once they enter aziel-runtime: a payload that must be refused or admitted. AZPIPE is the admission path.

FoldLock is the suppression hop. StaticClock-style freeze is the hash hop. FragGate is the refuse/door hop. SweepGate is the airlock. ChainLock is entry. Toolkits run last.

The hop list is the architecture. Changing product names does not change the list.

2. What it is not

- Not zip and not encryption. FoldLock is tether-word suppression on UTF-8.
- Not a TETH-1 completeness claim while `foldlock.py` is absent. Current hop is `fld3-wire`: allowlisted origins plus block-keys.
- Not a public Node Gate panel.
- Not QNM launch. The envelope reserves two bridges, isolate, and no body pull. Cells of 25 are specified elsewhere. This pipe does not join a cell.
- Not a substitute for host AV, sandboxing, or EmbryoLock wipe.

3. Hops

Inbound:

`frag → sweep → fold → static → fold → entry → frag → toolkits`

Outbound:

`toolkits → frag → fold → static → fold → sweep → frag`

frag (first). DecisionGATE flags D/E/I/G/R. Empty or ungrounded inbound refuses before SweepGate spends work. FragGate remains the kernel. This hop does not replace FragGate.

sweep. SweepGate airlock. Poison marks, malware-class structural marks, block-keys, off-origin URLs. Isolate and refuse. Do not merge. See SG-WP-0.1.

fold (first). FoldLock FLD3. If `foldlock.py` sits beside `azpipe.py`, load `suppress / fold / tether_suppress`. Otherwise `fld3-wire`: non-allowlisted `http(s)` becomes `[FLD3:url]`; keys named `password`, `private_key`, `secret`,

ssn, legal_name, home_address become [FLD3:block].

Allowlist: <https://www.azielcorpuslibrary.net>, <https://godlock.uk>, <https://www.azieleliab.com>, <https://aziel-runtime.vibelock.workers.dev>.

static. Canonical JSON (sorted keys, tight separators). Body SHA-256. Fact SHA-256. Pinned UTC timestamp. The freeze is what later hops hash. Volatile paths do not belong in the frozen body.

fold (second). Defense in depth. Tethers that appear only after canonicalization get the same suppression.

entry. ChainLock. Stamp, interact, recall, seal. The payload is now a fact-bearing card.

frag (second). Door before toolkits. Grounded claim or refuse. Toolkits do not run on an ungrounded envelope.

toolkits. Named MCP tools only.

Outbound is the reverse so a dirty toolkit result is swept before it rides a QNM bridge.

4. Envelope

```
{
  magic: "FLD3",
  v: "AZPIPE-0.2",
  dir: "in" | "out",
  path: [...hops],
  ok: true | false,
  h: sha256(inner),
  fh: sha256(clipped fact),
  isolate: {in, out},
  gates: {D,E,I,G,R,refuse},
  sweep: {airlock, hits, isolate, pull},
  foldlock: bool,
  teth: "foldlock.py" | "fld3-wire",
  qnm: {ready, pull:false, bridges:2, isolate:true},
  inner: {entry, toolkit, static}
}
```

Refuse envelopes stop at the hop that closed. They do not carry inner into memory.

Every ChainLock emit card also carries a thin pipe:{v,magic,path,h,fh,iso,qnm} so MCP output stays under 4096 bytes without dumping the full envelope. Full envelope: `chainlock.py pipe --dir in|out --json '{...}'`. Architecture: `pipe --arch`.

5. Isolate

Both directions isolate on the fact hash fh, not the whole stamp dump. Body hash remains on the freeze for byte integrity. QNM cites fh. Remote body does not enter STM.

Two payloads with the same fact text share fh. Two payloads with the same fact text and different wrapping URLs do not share body hash after fold, because off-origin URLs fold to a mark. That is intended.

6. Why this order

Frag first so SweepGate does not chew empty noise.

Sweep before fold so poison never becomes a folded "clean" string.

Fold before static so the freeze hashes the folded bytes.

Second fold after static so canonicalization cannot reintroduce a tether.

Entry only after the airlock is open.

Second frag so toolkits cannot run on an ungrounded card.

Outbound sweep so the mesh never ingests a toolkit leak.

A shorter list (frag–sweep–entry–frag) drops FoldLock and the freeze. That is cheaper and weaker. Long-term QNM needs both.

7. QNM slot

bridges=2, pull=false, isolate=true, sweep-before-ingest. Poison drops the tether. Tamper isolates from chain. This paper does not claim a live fleet or a qubit.

8. Cap

f1d3-wire is not TETH-1. Drop `foldlock.py` next to `azpipe.py` to switch lexicons without changing hops.

Companion papers: CL-WP-0.4, SG-WP-0.1, LS-WP-0.1.

SweepGate

Airlock, anti-poison, structural malware-class sweep

SG-WP-0.1 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Not a commercial antivirus product. Runtime: `sweepgate.py` · Version: SG-0.1

0. Sentence

A node mesh cannot treat inbound bytes as memory until they pass an airlock. SweepGate is that airlock. Isolate. Do not merge.

1. Claim

Public surfaces attract scrapers, prompt-injection, and drive-by payloads. GodLock already refuses poison on a 777-second class interval and cites instead of reanswering. SweepGate generalizes that refuse onto the AZPIPE hop list so every Worker, library call, and MCP envelope hits the same sieve before ChainLock entry.

The sieve is structural. It looks at the bytes. It does not claim vendor detection names, heuristic completeness, or host protection.

2. What it is not

- Not Windows Defender, ClamAV, or a Cloudflare WAF product.
- Not FoldLock. FoldLock suppresses tethers. SweepGate decides whether the envelope is admitted.
- Not FragGate. FragGate is the grounded-claim kernel. SweepGate is the airlock in front of it.
- Not a public Node Gate panel and not an IP allow/block UI. Hits are server-side. Gate config stays in `gate_config.json`.
- Not a promise that malware cannot run on the operator's laptop.

3. Classes

Poison. Marks such as inject-payload, jailbreak-ignore, exfiltrate, and explicit poison tokens. These are APG cousins. A hit isolates. The mesh records `rel=quarantine`. STM does not ingest the body.

Airlock-black. Keys and leftovers named `password`, `private_key`, `secret`, `legal_name`, `home_address`. Closed airlock. The runtime does not want those strings in cards or in a Worker log.

Off-origin. `http(s)` URLs that do not start with the allowlist (`azielcorpuslibrary.net`, `godlock.uk`, `azieleliab.com`, `aziel-runtime`). FoldLock will later mark them `[FLD3:url]`. SweepGate can isolate first when the envelope is inbound and untrusted.

Malware-class. Structural marks only:

- `script` tags
- `eval(`
- `PowerShell -enc`
- `cmd.exe`

- MZ / base64 MZ headers
- /bin/sh and rm -rf /
- dropper / meterpreter tokens

A hit is hits: ["malware-class"], airlock: closed, refuse: sweep-isolate. It is not a CVE name.

4. Wire

```
{
  v: "SG-0.1",
  ok: true | false,
  airlock: "open" | "closed",
  hits: [...],
  isolate: bool,
  pull: false,
  qnm: "sweep-before-ingest",
  h: sha256(raw)[:32]
}
```

AZPIPE inbound calls SweepGate after first frag. If isolate is true the envelope returns immediately. No fold. No static. No entry. No toolkit.

AZPIPE outbound calls SweepGate after the toolkit result is folded and frozen, so a dirty tool cannot ride a bridge.

5. Mesh

Sweep runs before QNM cell ingest. Cite the hash. Drop the tether. Quarantine stays on the mesh chain. Two rotating bridges, when live, only forward envelopes with airlock: open and pull: false.

Operator traffic is not a special public button. Operator bypass, if any, is a server-side token in gate config. Same rule as cost control on the Worker: no Node Gate panel.

6. Relation to cost and poison

The Cloudflare bill that motivated rate limits is a cousin problem. Scrapers that walk every software tab also walk KV. SweepGate does not replace per-visitor request caps. It stops the class of payload that should never become a card even if the visitor is under quota.

7. Cap

This is a structural sieve on the pipe. It does not replace host AV, sandboxing, Cloudflare WAF, or EmbryoLock wipe policy. Expanding the mark list is an operator act and must be stamped on learn.

Companion papers: CL-WP-0.4, AP-WP-0.2, LS-WP-0.1.

LOCKSET

ChainLock × *TemporalLock* × *GodLock*

LS-WP-0.1 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Cite, do not impersonate the public ledger. Command: `chainlock.py seal` · Rule: one member fail fails the set

0. Sentence

Public verification of local memory is a three-body lattice, not a single hash. If one member fails, the set is false.

1. Claim

ChainLock holds tips. TemporalLock writes a neutral receipt. GodLock is the public cite. Those three must agree on one canonical object called LOCKSET. Agreement is mechanical: hashes match, tips have not drifted, both companion blocks exist.

This is the same instinct as GodLock citation-over-reanswer and as TemporalLock's refusal to narrate. LOCKSET just binds them so a reader cannot treat a local vault as "verified publicly" when the public cite is missing.

2. What it is not

- Not a GodLock write path. `seal` cites `https://godlock.uk`. The operator posts the lockset SHA-256.
- Not a TemporalLock rewrite. The receipt is a new file under `vault/receipts/`. Old receipts stay.
- Not encryption. LOCKSET does not seal privileged plaintext. It seals hashes of public-plane cards.
- Not a live QNM membership proof.

3. Members

Each live chain contributes:

- `c` chain name
- `id` tip stamp id
- `h` tip stamp_sha256
- `fh` SHA-256 of the clipped tip fact

Chains with no stamps are omitted. Empty vault cannot seal.

Temporal block:

- `kind`: TemporalLock
- `ts`: UTC
- `note`: "Neutral receipt. No narrative. No authority."
- `n`: member count

GodLock block:

- `node`: godlock

- url: `https://godlock.uk`
- cite: `public-verify`
- pull: `false`
- isolate: `true`

`lockset_sha256` is SHA-256 of the canonical JSON of those fields with the hash field omitted. The file `vault/receipts/LOCKSET.json` is the live set. A copy `vault/receipts/TL-<prefix>.json` is the TemporalLock receipt. A mesh stamp records the cite. After that mesh append, members are recomputed so the mesh tip is inside the set.

4. Verify

`chainlock.py verify` now returns `lattice` as well as `per-chain ok`.

Fail-closed conditions:

- any chain prev break, body-hash miss, stamp-hash miss, or SSH-fail
- LOCKSET file missing after a seal has been made (sealed vaults only; unsealed vaults verify chains alone)
- LOCKSET file unreadable
- stored `lockset_sha256` $\neq$ recomputed hash
- any stored tip `h` $\neq$ live tip `h` (tip drift)
- missing `godlock` block
- missing `temporal` block

Any one of those sets `ok: false` and `lattice: false`. Breaks are listed. History is not rewritten.

5. Drift

Interact, append, library-sync, mesh-cite, and compact all move tips. That is correct. The set is a snapshot. After material change, seal again. An old TL receipt remains as history. The live file is always `LOCKSET.json`.

6. Public path

Post the lockset SHA-256 on GodLock. Do not upload stamp bodies. Citation over reanswer. 777s-class anti-poison on the public node is unchanged.

A reader who has the vault and the public cite can check: recompute the lockset hash, confirm the GodLock entry names that hash, confirm live tips still match. If GodLock is dark, the local receipt still exists. That is site-fail resilience, not public verification.

7. Cap

Seal cites. It does not write the GodLock ledger from this runtime. Do not print legal name or home on the receipt.

Companion papers: CL-WP-0.4, AP-WP-0.2, SG-WP-0.1.

Rate Limit and Donation Plan

Keep aziel-runtime public. Stop paying for KV walks. Operator uncapped.

RL-WP-0.1 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification and cost plan. Not a fleet-completeness claim. Evidence window: Cloudflare billable usage, 2–9 September 2026 (8 of 30 cycle days).

0. Sentence

Users are cheap. The catalog walk is not. Limit visitors on the Worker. Do not limit the operator. Do not hide the runtime.

1. What the bill said

Observed 2–9 September 2026:

- Total cost: \$142.26
- Projected cycle cost: \$533.47 (2 September – 1 October, 8 of 30 days)
- Average daily: \$17.78
- KV Read Operations: 271.91M — **\$131.00**
- KV List Operations: 1.33M — \$5.00
- KV Write Operations: 1.37M — \$5.00
- Workers CPU ms: 92.36M — \$1.26
- Workers Standard Requests: 1.42M — inside the 10M included
- D1 rows read: 5.44B — inside the 25B included (\$0)
- KV / D1 storage: 0 GB-month

Workers KV paid plan: first 10 million reads per month included, then \$0.50 per million. $\$131 / \$0.50 = 262$ million billable reads. Plus the 10 million included ≈ 272 million. That matches the dashboard.

1.42 million Worker requests against 271.91 million KV reads is about **191 KV operations per request**. The public is not hitting the Worker 272 million times. Something on the hot path lists or gets that many keys per page.

CPU and request counts are fine. D1 is fine. Storage is fine. The bill is almost entirely KV reads.

This paper does not name a private account id, a home address, or an IP.

2. Claim

A packed catalog key, an edge cache, a per-visitor request cap, and a server-side operator token cut billed KV reads by an order of magnitude without shrinking the public surface and without rate-limiting the operator.

Donation is a static tab. It does not touch KV.

3. What it is not

- Not a public Node Gate panel.

- Not an IP allow/block UI on azieleliab.com, godlock.uk, or the library.
- Not a shutdown of aziel-runtime.
- Not a claim that Cloudflare WAF replaces this plan.
- Not invented wallet addresses. Addresses are an operator paste at publish time.
- Not a GodLock ledger write.

4. Root cause

Hot-path patterns that produce 191 reads per request:

- `KV.list()` to build the software / FragGate / library index on every hit
- key-by-key `KV.get` of every catalog row instead of one packed JSON
- `cacheTtl` omitted, so every get bills
- dashboard and wrangler list/get counting as billable KV
- scrapers walking every software tab, each tab repeating the list

Fix the walk first. Rate limit second. Donation third.

5. Plan, in order

Step 1 — Pack the catalog. One key: `catalog:v1` (or `software.json` plus `library-index.json`). Shape is the AZL-VOL door list already specified: slug, title, one-liner, hash, href. No PDF bodies. `list()` leaves the hot path.

Step 2 — Cache. `env.KV.get(key, { cacheTtl: 3600 })`. Public JSON responses carry `Cache-Control: public, s-maxage=300, stale-while-revalidate=3600`. Repeat visitors hit the Cloudflare cache, not KV. Static software tabs move to assets where possible. Static asset requests are free.

Step 3 — Cap KV work inside one invocation. A request may perform at most 30 KV operations. If a handler would exceed that, return the packed catalog plus truncated. Do not walk the rest.

Step 4 — Per-visitor limit, server-side. Identify a visitor as SHA-256 of CF-Connecting-IP plus optional cookie. No public “who am I” control.

Default bucket (everyone except operator):

- 60 requests / minute
- 600 requests / hour
- 5,000 requests / day

Soft exceed: HTTP 429, `Retry-After: 30`, body is the last cached catalog so the site still “runs.” Hard isolate: SweepGate poison / scraper class only. Cite, drop tether, do not merge.

Step 5 — Operator exclude. A secret request header or Cloudflare Access service token, stored only in `gate_config.json` on the Worker. Same token the standing rules already keep off the public surface. Operator traffic skips the bucket. Operator still uses packed catalog and cache. Operator is not a reason to `list()` KV.

Step 6 — Cycle cap. Cloudflare budget alert at \$80–\$100 for the billing cycle. A scrape cannot run the projection to \$533.

Step 7 — Measure. After ship: KV reads / Worker request should fall from ~191 toward 1–3 on catalog pages and toward 1 on cached hits. If it does not, the packed key is not on the hot path yet.

6. Visitor still has the runtime

A limited visitor receives:

- the last good catalog
- FragGate / search / mesh within quota
- SweepGate airlock (unchanged)
- ChainLock local memory on their machine, which does not bill this Worker

They lose only unbounded crawl. They do not lose the doctrine, the library cards they already fetched, or the local vault.

QNM launch does not change this. Bridges still `pull=false`. Sweep still runs before ingest.

7. Donation tab

Add a Donate strip on azieleliab.com, the library, and godlock.uk.

Rules:

- Static page or static island. No Worker. No KV.
- Networks the operator already controls: Bitcoin, Lightning, Ethereum, Solana. Paste addresses at publish. Do not invent them in this paper.
- Copy button plus a one-line receipt note: date, network, tx hash if the donor sends it. Receipt is optional and public-plane only.
- No legal name. No home. No “support my custody case” copy on the public tab.
- Do not route donations through Workers KV, D1, or Durable Objects.

The tab does not lower the KV bill. It funds the cycle cap so the public surface can stay up.

8. Wire sketch

IN: frag → sweep → [bucket] → fold → static → fold → entry → frag → toolkits

[bucket] sits after SweepGate. Poison never spends a rate token that matters. Operator token short-circuits the bucket only.

Refuse classes:

- sweep-isolate — payload
- rate-soft — 429 + cached catalog
- rate-hard — scraper isolate
- kv-cap — truncated packed response

None of those are a public panel.

9. Success

- KV reads / request ≤ 3 on catalog pages
- Monthly KV reads near or under the 10 million included, or low enough that \$0.50/million is noise
- Projected cycle cost back under \$50 unless traffic is real humans at scale
- Operator verify / seal / library-sync still works at full depth
- No Node Gate ships

10. Cap

This paper is a plan plus a doctrine cap. It is not the Worker patch itself. The patch lives in aziel-runtime and stays server-side.

Companion papers: CL-WP-0.4, AP-WP-0.2, SG-WP-0.1, LS-WP-0.1.